

Kayle Megginson

Bend, OR — 503-562-9029 — Kaylemegginson@gmail.com — LinkedIn — Portfolio

Objective

Recent Mechanical Engineering and Computer Science graduate with experience in systems modeling, automation, data analysis, and technical problem-solving. Eager to apply engineering and software skills to develop innovative solutions while continuing to learn and grow in a collaborative environment.

Education

Oregon State University Jun. 2026

M.Eng. Mechanical Engineering (Dynamics & Controls)

Willamette University - 3.93 GPA Aug. 2025

M.S. Computer Science — B.S. Computer Science & B.S. Data Science — Applied Mathematics Minor

Technical Skills

Languages: Python, C, JavaScript, CSS, MATLAB, R, SQL, C++

Tools: Git, Docker, REST APIs, Arduino, Flutter, PostgreSQL, MySQL, SolidWorks, AutoCAD

Domains: ML, product design, data analysis, optimization, system modeling, backend systems

Work Experience

Black Butte Ranch, Sisters, OR May 2025 – Present

Wedding Server

- Made a tool that parsed BEO documents and generated buffet cards, saving 15–20 minutes per event
- Coordinated event setups for up to 210 guests, optimizing layout for efficiency and flow
- Managed 3–5 service stations while maintaining inventory and service quality
- Streamlined cleanup workflow, reducing setup time by 20%

Santiam Jr/Sr High School, Mill City, OR Sep. 2021 – Jun. 2022

Mechanic

- Diagnosed issues in 10+ donated vehicles using structured troubleshooting methods
- Repaired mechanical and electrical systems, increasing salvage value by an estimated 40%
- Managed \$4,000 in parts inventory and shipment tracking

Projects

- Used Arduino accelerometer data to compare spring-mass models with FFT results and hardware behavior.
- Built and deployed a cloud-based financial app using Flutter, AWS, MySQL, and REST APIs
- Solved engineering optimization problems using MATLAB to improve performance under constraints
- Built a path algorithm to optimize routing across a flight network, reducing average path cost by 76 %
- Implemented QPSK communication system with full transmit/receive pipeline
- Performed survival analysis and statistical modeling on large datasets

Engineering & Data Experience

- Used optimization methods to improve system performance, cost, and design tradeoffs
- Modeled dynamic systems using linear systems theory, structural dynamics, and control-oriented analysis
- Applied digital signal processing (filtering, time/frequency analysis) for system modeling
- Designed engineering solutions using CAD tools with focus on manufacturability and constraints
- Evaluated material behavior and failure modes using fracture mechanics and composites principles

Leadership & Collaboration

- Led multidisciplinary teams in ambiguous problem-solving environments
- Communicated technical insights through data-driven presentations